PROJECT DOCUMENTATION

Household Electric Power Consumption Forecasting using RNN and LSTM

GitHub Repository:
https://github.com/priyanithi876-ship-it/Assignment-10-Household-Electric-Consumption-Forecasting-using-RNN-and-LSTM

Notebook: ASSIGNMENT_10_RNN_MODEL_TIME_SERIES_PREDICTION.ipynb

1. Introduction

This project studies household electric power consumption as a time series and explores recurrent neural network approaches for forecasting. RNN and LSTM models are designed to learn patterns across sequential observations.

2. Problem Statement

Electricity consumption changes over time. The project uses historical household consumption measurements to learn temporal patterns and predict future consumption values for the selected target variable.

3. Objectives

Load and inspect the electricity-consumption dataset.

Clean missing or invalid observations and prepare time-related fields.

Transform the data into sequences suitable for recurrent models.

Train an RNN and/or LSTM model as implemented in the notebook.

Evaluate forecasts on held-out time periods and visualize actual versus predicted values.

4. Dataset Description

The project references the Individual Household Electric Power Consumption dataset hosted by the UCI Machine Learning Repository.

Dataset page: https://archive.ics.uci.edu/dataset/235/individual+household+electric+power+consumption

Archive URL used in the notebook:
https://archive.ics.uci.edu/static/public/235/individual+household+electric+power+consumption.zip

The dataset contains time-stamped household electric power measurements. Consult the dataset documentation and notebook to confirm the exact features, target column, date range, and sampling frequency used.

5. Tools and Technologies

Python; Jupyter Notebook or Google Colab; pandas; NumPy; Matplotlib; scikit-learn; TensorFlow/Keras if used by the notebook.

6. Methodology

- Step 1 — Data loading: load the downloaded dataset into a DataFrame.
- Step 2 — Data inspection: review columns, data types, missing values, and basic statistics.
- Step 3 — Data cleaning: parse timestamps and handle missing or invalid readings using a documented approach.
- Step 4 — Feature and target selection: choose the input variables and forecast target used in the notebook.
- Step 5 — Train/test split: split observations in chronological order to preserve the time-series sequence.
- Step 6 — Scaling: fit preprocessing transformations on training data only, then apply them to the test data.
- Step 7 — Sequence generation: create rolling input windows and corresponding prediction targets.
- Step 8 — Model training: train the RNN/LSTM architecture defined in the notebook.
- Step 9 — Prediction and evaluation: generate forecasts and calculate the metrics implemented in the notebook.
- Step 10 — Visualization: plot actual and predicted values to review model behaviour.

7. Model Overview

RNNs process sequential information by carrying information from earlier time steps. LSTMs are a type of recurrent network that use gates and a memory state to learn longer-range dependencies. The actual layer configuration, input window length, optimizer, loss function, and training settings should be copied from the final executed notebook.

8. Evaluation Metrics

MAE (Mean Absolute Error) measures the average absolute difference between actual and predicted values.

MSE (Mean Squared Error) averages squared prediction errors and penalizes larger errors more heavily.

RMSE (Root Mean Squared Error) is the square root of MSE and is expressed in the target variable's units.

Only include metrics that were calculated by the notebook, and report their actual test-set values.

9. Results and Discussion

Insert the actual evaluation values and the final actual-versus-predicted plot after running the notebook. Do not report placeholder or assumed scores.

- MAE: To be filled from notebook output.
- MSE: To be filled from notebook output.
- RMSE: To be filled from notebook output.

Discuss whether predictions follow the main trend, where errors are larger, and how the model compares with a simple baseline if one is available.

10. Repository Structure

Suggested structure:

Assignment-10-Household-Electric-Consumption-Forecasting-using-RNN-and-LSTM/

README.md

requirements.txt

RNN Model - Time Series Prediction/

ASSIGNMENT_10_RNN_MODEL_TIME_SERIES_PREDICTION.ipynb

README.md

docs/

Project_Documentation.pdf

Keep names aligned with the files actually uploaded. GitHub does not keep empty folders; add a file inside each folder.

11. How to Run the Project

Open the notebook in Google Colab or Jupyter Notebook.

Install required dependencies if they are not already available.

Check the dataset download or file path in the data-loading cell.

Run the notebook cells in order and verify that each stage completes successfully.

Update the results section with the metrics and plots produced by the final run.

12. Limitations

Forecast quality depends on data quality, selected features, sequence length, model settings, and the time period used for testing.

Randomly shuffling time-series observations can cause leakage; use a chronological split.

Preprocessing such as scaling should be fitted on training data only.

Performance on one test period may not represent performance during all seasons or usage patterns.

13. Future Enhancements

Compare RNN and LSTM models under the same data split and evaluation protocol.

Test a simple baseline and additional sequence lengths.

Tune model hyperparameters systematically.

Evaluate forecasts across multiple time windows.

Save the model and preprocessing objects if the project will later be deployed.

14. Conclusion

This project demonstrates a workflow for preparing household electricity-consumption time-series data and applying recurrent neural networks for forecasting. The final conclusion should be updated after reviewing the notebook's real evaluation metrics and plots.

15. References

UCI Machine Learning Repository — Individual Household Electric Power Consumption: https://archive.ics.uci.edu/dataset/235/individual+household+electric+power+consumption

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TensorFlow/Keras documentation, if TensorFlow/Keras is used: https://www.tensorflow.org/